

Local-Stack Topology — Deployment Architecture

Status: descriptive architecture reference. The Android client, the lava-api-go service, the root docker-compose.yml, and mDNS discovery are **in-tree and operational**. The **Jackett + FlareSolverr sidecar** is at **Phase 1** — the Torznab Go client + a compose fragment exist and are tested, but **no container has been booted** and **no lava-api-go handler routes through Jackett yet** (marked **NOT YET WIRED** below). Per HelixConstitution §11.4.6 (no-guessing vocabulary), the implemented-vs-pending split is read from source, not inferred.

Companion docs: [docs/qa/jackett-local-stack-research.md](#) (research dossier, every external claim cited), [docs/qa/jackett-local-stack-implementation.md](#) (Phase 1 implementation + verbatim test output), [docs/LOCAL_NETWORK_DISCOVERY.md](#) (mDNS), [docs/architecture/tracker-sdk.md](#) (client-side SDK).

1. The whole stack in one diagram

The load-bearing invariant: **the Android app only ever talks to lava-api-go**. Jackett and FlareSolverr are server-side sidecars on the internal lava-net bridge; they are never advertised on the LAN and never reachable from the device. Credentials and the Jackett api_key stay server-side (§6.H).

graph TD

```
APP["Android app<br/>digital.vasic.lava.client(.dev)<br/>core:data LocalNetworkDiscoveryService"]
```

```
subgraph HOST["host machine (rootless podman/docker via Containers submodule)"]
```

```
API["lava-api-go<br/>:8443 HTTP/3 + HTTP/2 TLS<br/>network_mode: host<br/>Lava-Auth HMAC gate"]
```

```
subgraph BRIDGE["lava-net (bridge network)"]
```

```
PG[("lava-postgres<br/>:5432 (127.0.0.1:8432 host)")]
```

```
JK["lava-jackett<br/>:9117 Torznab<br/>expose-only,<br/>NOT advertised<br/>NOT YET WIRED"]
```

```

        FS["lava-flaresolverr<br/>:8191 headless Chromium<br/>
>profile=cloudflare<br/>NOT YET WIRED"]
    end
end

APP -->|"mDNS _lava-api._tcp<br/>then TLS + Lava-Auth"| API
API -->|"storage.Storage (Postgres adapter)"| PG
API -->|"Torznab HTTP (server-side, holds api_key)<br/>NOT
YET WIRED"| JK
JK -->|"Cloudflare challenge solve<br/>(only when
profile=cloudflare)"| FS

classDef pending stroke-dasharray: 5 5;
class JK,FS pending;

```

Dashed boxes/edges = **NOT YET WIRED** (Phase 1: client + compose fragment exist, no boot, no handler). Solid = in-tree and operational.

Grounding:

- network_mode: host for lava-api-go, lava-net bridge for Postgres — docker-compose.yml (top comment: *"lava-api-go uses network_mode: host so JmDNS / mDNS advertisements reach the LAN; Postgres + observability containers use the lava-net bridge"*).
 - mDNS _lava-api._tcp, :8443 TLS, host-net — confirmed in docs/qa/jackett-local-stack-research.md §0 against lava-api-go/cmd/lava-api-go/main.go + internal/config/config.go.
 - Postgres service name lava-postgres, host port 127.0.0.1:8432:5432 — docker-compose.yml.
 - Jackett / FlareSolverr compose fragment — tools/lava-containers/docker-compose.jackett.yml (per docs/qa/jackett-local-stack-implementation.md Part 2).
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2. The app sees ONE endpoint

The client's entire view of the backend is a single _lava-api._tcp (or _lava-api-dev._tcp for dev) mDNS service that resolves to one Endpoint.GoApi(host, port). Jackett is invisible to the app by design — it is a server-side capability that lava-api-go proxies over Torznab.

sequenceDiagram

```

actor User
participant APP as Android app
participant NSD as NsdManager / LocalNetworkDiscoveryService
participant API as lava-api-go (:8443)
participant JK as lava-jackett (:9117)

```

```

User->>APP: open Settings
APP->>NSD: discover _lava-api._tcp (5s scan)
NSD-->>APP: Endpoint.GoApi(host, port)
User->>APP: search "ubuntu"
APP->>API: GET /v1/search (TLS + Lava-Auth HMAC)
Note over API,JK: NOT YET WIRED – when a Torznab provider
lands:
  API->>JK: GET /api/v2.0/indexers/<id>/results/torznab/api?
  apikey=...&t=search&q=ubuntu
  JK-->>API: RSS/Torznab XML (items + enclosures)
  API-->>APP: JSON results (app never sees Jackett)
  APP-->>User: render rows

```

Security consequences (from research dossier §7 + implementation §"Ports exposed to the LAN"):

- Neither :9117 (Jackett) nor :8191 (FlareSolverr) is published — they use expose: (intra-bridge only). Only lava-api-go:8443 is LAN-visible.
- The discovery + auth model is unchanged: one endpoint, existing TLS + HMAC. /health and /ready answer without a credential so discovery probes work; any data endpoint returns 401 without the correct Lava-Auth key (see docs/LOCAL_NETWORK_DISCOVERY.md "Authentication when connecting").

3. mDNS discovery flow (in-tree, operational)

```

graph LR
  ADV["lava-api-go advertiser<br/>_lava-api._tcp<br/>(LAVA_API_MDNS_TYPE)"]
  SVC["LocalNetworkDiscoveryService<br/>core:data – NsdManager + callbackFlow"]
  UC["DiscoverLocalEndpointsUseCase<br/>core:domain – firstOrNull + timeout"]
  VM["MenuViewModel<br/>feature:menu"]
  UI["MenuScreen bottom sheet<br/>feature:menu"]

  ADV -->|"multicast 5353/UDP"| SVC
  SVC -->|"Flow<DiscoveredEndpoint>"| UC
  UC -->|"DiscoverLocalEndpointsResult"| VM
  VM -->|"MenuSideEffect.OpenConnectionSettings"| UI

```

Service types the client resolves (verbatim from docs/LOCAL_NETWORK_DISCOVERY.md):

Source	Service type	TXT records
Host server (release)	_lava-api._tcp	engine=go, platform=server (or absent)
Host server (dev)	_lava-api-dev._tcp	engine=go-dev
On-device app (release)	_lava-api._tcp	engine=go, platform=android, storage=sqlite
On-device app (dev)	_lava-api-dev._tcp	engine=go-dev, platform=android, storage=sqlite

LocalNetworkDiscoveryServiceImpl resolves these to Endpoint.GoApi(host, port) via the authoritative TXT engine attribute (falling back to the service type). The on-device **advertiser** (a phone/tablet registering itself via NsdManager) is **PENDING Phase D** — see docs/ON_DEVICE_API.md. Client-side discovery is in-tree and unchanged.

4. Jackett sidcar — Phase 1 detail (NOT YET WIRED into a handler)

4.1 What exists today

```
graph TD
    subgraph "lava-api-go/internal/jackett (in-tree, tested)"
        TZ["torznab.go<br/>encoding/xml decoder → Result struct"]
        CL["client.go<br/>Client: BuildSearchURL/BuildCapsURL<br/>Search, Download (302→magnet)"]
        TD["testdata/torznab_results.xml<br/>+ torznab_test.go (12 tests PASS)"]
    end
    FRAG["tools/lava-containers/docker-compose.jackett.yml<br/>compose fragment (parses clean, no boot)"]
    HANDLER["lava-api-go HTTP handler / provider route<br/>NOT IMPLEMENTED"]

    CL --> HANDLER
    FRAG -->|"defines lava-jackett + lava-flaresolverr"| HANDLER

    classDef pending stroke-dasharray: 5 5;
    class HANDLER pending;
```

Per docs/qa/jackett-local-stack-implementation.md:

- internal/jackett/torznab.go + client.go + torznab_test.go exist; the test suite reports ok digital.vasic.lava.apigo/internal/

jackett (12 tests PASS, verbatim output in the implementation doc).

- tools/lava-containers/docker-compose.jackett.yml is a **fragment** merged at run time with the root docker-compose.yml via -f; it parses clean (podman compose -f docker-compose.jackett.yml config --quiet) but **no container was booted**.
- **NOT IMPLEMENTED:** the lava-api-go HTTP handler / provider route that calls jackett.Client, the OpenAPI spec entry, indexer provisioning automation, and .env / .env.example keys.

4.2 The 302 → magnet handler (the correctness-critical bit)

Jackett answers a download link with **HTTP 302 whose Location is a magnet: URI**. A naive auto-following client tries to "GET magnet:" and fails. jackett.Client sets http.Client.CheckRedirect = http.ErrUseLastResponse (do **not** follow) and reads the Location header:

graph TD

```
DL["Download(ctx, item.DownloadURL)"]
Q{"enclosure already magnet?"}
SHORT["short-circuit → DownloadResult.Magnet<br/>(no HTTP round-trip)"]
HTTP["HTTP GET (CheckRedirect = ErrUseLastResponse)"]
R302{"302?"}
LOC{"Location starts with magnet:?"}
MAG["DownloadResult.Magnet (captured)"]
ERR["explicit error<br/>(do NOT follow non-magnet redirect)"]
BODY["HTTP 200 → DownloadResult.TorrentBytes + ContentType"]
```

```
DL --> Q
Q -->|yes| SHORT
Q -->|no| HTTP --> R302
R302 -->|yes| LOC
LOC -->|yes| MAG
LOC -->|no| ERR
R302 -->|no, 200| BODY
```

This serves the download-confirmation goal (a valid .torrent byte stream OR a real magnet) and closes the known kinozal/nmclub null-magnet gaps — and pairs with the client-side TorrentFileValidator / MagnetLinkValidator (core/common/.../torrent/, see [tracker-sdk.md §7](#)) so a download is provably real on both sides of the boundary.

4.3 FlareSolvrr — profile-gated

FlareSolvrr (headless Chromium, the RAM-heavy component) is gated behind a cloudflare compose profile so it is only up when a Cloudflare/DDoS-Guard indexer (e.g. IPTorrents) is enabled. For RuTracker/RuTor/NNMClub it stays off. Jackett points at it via a FlareSolvrr URL in Jackett's config; readiness is `POST :8191/v1 {"cmd":"sessions.list"} → 200` (no native /health). Source: research dossier §4.

4.4 Who holds the api_key

```
graph LR
    GEN["Jackett generates api_key on first run"]
    VOL["/config/ServerConfig.json<br/>(gitignored host volume<br/>LAVA_JACKETT_CONFIG_DIR)"]
    ENV["LAVA_JACKETT_API_KEY<br/>(env, populated by Containers glue)"]
    CFG["jackett.Config{ BaseURL, APIKey }<br/>in lava-api-go (server-side)"]
    DEV["Android device<br/>NEVER receives api_key"]

    GEN --> VOL
    VOL --> ENV
    ENV --> CFG
    CFG --> DEV

    GEN -. "§6.H: never sent" .-> DEV
```

The `api_key` is a §6.H secret: read server-side at startup, **never** hardcoded (§6.R), **never** sent to the device. Source: docs/qa/jackett-local-stack-implementation.md "Who holds the apikey".

5. Orchestration — the Containers submodule

The stack is brought up by `tools/lava-containers/cmd/lava-containers` (thin Lava-side glue), which auto-detects rootless podman/docker and merges the root `docker-compose.yml` with fragments. The Jackett fragment composes in as:

```
docker compose -f docker-compose.yml \
  -f tools/lava-containers/docker-compose.jackett.yml \
  --profile api-go --profile jackett config
```

Generic container-runtime concerns are owned by the `vasic-digital/Containers` submodule (submodules/containers/, pinned); the local CLI is thin glue per the Decoupled Reusable Architecture rule. **Do not bring docker-compose.yml up directly** — orchestration is owned by the CLI (`start.sh` / `stop.sh`).

Compose profiles (from `docker-compose.yml` header):

Profile	Brings up
api-go (default)	lava-postgres + lava-migrate + lava-api-go
observability	Prometheus + Loki + Promtail + Tempo + Grafana
dev-docs	Swagger UI mounting api/openapi.yaml
jackett (fragment)	lava-jackett (NOT YET WIRED)
cloudflare (fragment)	lava-flaresolverr (NOT YET WIRED)

6. Healthchecks (§6.B — "Up" is not "healthy")

`docker ps` reporting Up only means PID 1 is alive. Each service has an application-level probe:

Service	Probe	Why
lava-postgres	<code>pg_isready -U lava -d lava_api</code>	DB accepting connections (<code>docker-compose.yml</code>).
lava-api-go	Dockerfile HEALTHCHECK → healthprobe (real HTTP/3)	distroless, no <code>/bin/sh</code> ; §6.A contract test at <code>lava-api-go/tests/contract/healthcheck_contract_test.go</code> .
lava-jackett (planned)	Torznab <code>t=caps 200 + parseable <caps></code>	proves the Torznab surface is live AND <code>api_key</code> is valid — the real code path <code>lava-api-go</code> will consume (not a bare TCP probe).
lava-flaresolverr (planned)	<code>POST /v1 {"cmd":"sessions.list"} → 200</code>	no native <code>/health</code> ; a bare TCP-8191 would accept the Python proxy before Chromium is ready.

Any future Lava-owned healthprobe-style binary used for the Jackett probe must gain a §6.A contract test asserting its flag set, mirroring `healthcheck_contract_test.go` (research dossier §0 forensic anchor: the 569-consecutive-failing-healthcheck incident where the probe rejected a flag the compose file passed).

7. Open questions (carried from the research dossier)

These are operator/architecture decisions, not yet resolved (verbatim from docs/qa/jackett-local-stack-research.md §"OPEN QUESTIONS"):

- **Q1** — Jackett vs Prowlarr final call (Prowlarr only wins on provisioning automation; consumption surface is ~equal Torznab).
- **Q2** — Distribution mode for the Jackett image: reference `lscr.io/...` by digest vs. re-push into a Lava-owned registry (drives GPL-2.0 obligations).
- **Q3** — Indexer provisioning path: internal admin HTTP endpoints (version-coupled, undocumented) vs. `per-/config` JSON (undocumented schema).
- **Q4** — Jackett network mode: `lava-net` bridge (recommended) vs. `host-net`.
- **Q5** — FlareSolverr arm64 Chromium boot under the podman VM on Apple Silicon must be boot-verified before it is gate-eligible.

Last updated: 2026-06-08. Implemented-vs-pending split verified against the working tree + the qa/Jackett docs.